

NARWC Database Project

Progress Update

July 2026

VISUAL PLACEHOLDER

Title visual — NARWC/NEAQ logo or a representative survey photo

Project Scope

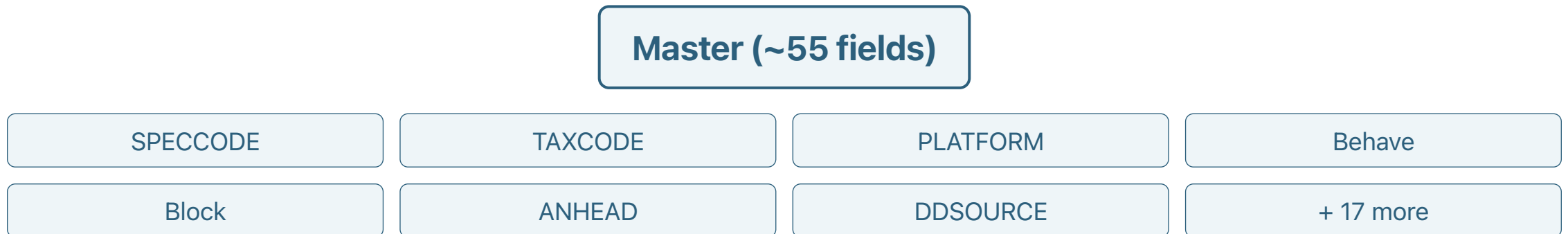
- Migrate the legacy NARWC database into a modern, validated SQL database
- Build tools to keep curating and ingesting new survey data going forward
- Target: migration complete and curation tools ready by end of August 2026

Legacy Database Migration to SQL Server

- Rebuilt the full legacy structure as a modern SQL schema
- Automated pipeline: extract, validate, upload — run against the entire historical dataset
- Every record is checked before it's allowed in; nothing is silently changed

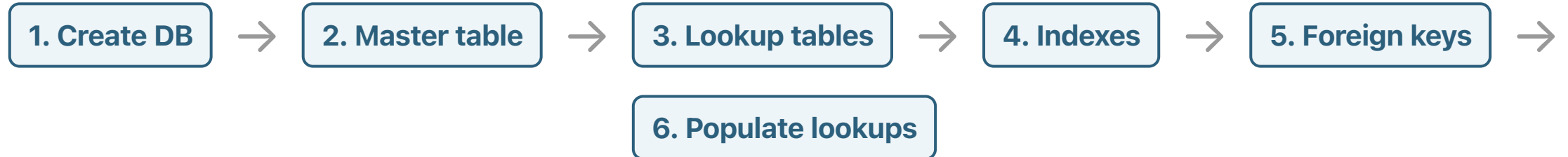
The Database Schema

- One central table (~55 fields); **24 lookup tables** constrain species, behavior, platform, and more
- **35 foreign key constraints** — the database itself rejects unrecognized codes, even if something slips past validation



Setting Up a New Database

Run once, in order, against a fresh SQL Server instance:



Scripts are idempotent and credential-free — connect externally first.

The Data Pipeline

Two ways surveys enter the database — only the first step differs:

Migration (one-time)

Extract

Routine (ongoing)

Convert



Validate



Upload

The database is a second line of defense via foreign key constraints.

How It's Built

- One shared validation and upload process, no matter where data came from
- Adding a new contributor is a small, isolated piece of work
- Backed by an automated test suite, so changes can't silently break what already works

Data Curation Library & Tools

- MATLAB library for importing and curating surveys, built to extend to new providers
- Automated handling for known data-entry patterns
- Change tracking and quality-control reporting throughout

Validation: What Gets Checked

Every survey is checked against 9 rule categories:

- Required fields, coordinates, dates and times
- Species, taxonomic, and behavior codes
- Environmental conditions and Beaufort sea state
- Cross-reference integrity against lookup tables

Curators can tune thresholds themselves — no code change required.

Validation In Action — Real Errors

```
[ERROR] TAXCODE: TAXCODE mismatch: got 2, expected 1 for SPECCODE "RIWH" (rows 2, EVENTNO=42)
[ERROR] SPECCODE: SPECCODE "RIWHT" exceeds 4 characters (rows 1, EVENTNO=5)
[ERROR] TAXCODE: TAXCODE is required for sighting records (rows 51)
[ERROR] GLARER: GLARER contains invalid value(s) not in GLARE lookup table: 9 (+427 more rows)
```

Single-record issues show the exact EVENTNO; issues spanning many records show a count instead.

Validation In Action — Real Warnings

```
[WARNING] BEHAV: Calf-associated behavior(s) 40 recorded but no calf present (rows 1, EVENTNO=10)  
[WARNING] LONG_DD: Longitude outside typical survey area [-85.0, -40.0] (rows 1, EVENTNO=10)  
[WARNING] LAT_DD: Latitude outside typical survey area [20.0, 55.0] (rows 22, EVENTNO=380)  
[WARNING] NUMCALF: NUMCALF (2) is more than half of NUMBER (3) – verify count (rows 45, EVENTNO=370)
```

Warnings never block silently — a curator reviews each one and either fixes the data or records why it's expected.

Known, Automatic Corrections

These reproduce corrections Bob has already made in the official database — so migrated data matches what he's already hand-corrected.

```
=== Known Fixes Applied ===  
PHOTOS = 0 -> 1 (sighting rows): 3 rows across 2 survey(s)  
STRIP > 16 -> NULL (NEAq 2021): 0 rows  
BEAUFORT = 99 -> NULL: 0 rows
```

Every fix is logged with a row count in that run's error log.

How To Run It — Getting Set Up

```
startup % add paths, check toolboxes, create data dirs  
scripts/setup/test_connection % verify the database connection
```

Checks config, connection, and sample data with a pass/fail checklist.

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Console output — screenshot of the startup/connection-check checklist

How To Run It — Convert a Contributor's Batch

```
convert_contributor_batch('neaq', 'NEAQFormat')
```

- Converts a contributor's raw files into standard-format survey files, ready to upload
- New contributors just need one small parser, written once

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Console output — screenshot of a conversion run

How To Run It — Upload & Validate

```
upload_contributor_batch('Config', load_config('routine'))
```

- Validates and uploads everything that passes
- Anything that doesn't goes to a review queue with a log explaining why

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Console output — screenshot of an upload run

The Override System — Acknowledging One Warning

Curators record a decision — name, date, reason — in a version-controlled file:

```
fileid,eventno,field,rule_id,acknowledged_by,acknowledged_date,reason  
f098027,42,LAT_DD,coordinate_rules.outside_survey_lat,RS,2026-06-25,Opportunistic sighting near shore
```

Re-run the upload to pick up the acknowledgement.

The Override System — Acknowledging a Whole Survey

Leave EVENTNO blank to acknowledge every matching warning in a survey at once:

```
fileid,eventno,field,rule_id,acknowledged_by,acknowledged_date,reason  
p905169G,,BEHAV,behavioral_rules.calf_behavior_no_calf,RS,2026-06-25,All 93 calf-behavior events reviewed; survey predates current coding convention
```

Exact per-row matches win; per-survey is the fallback. Committed to git, so the whole team shares the same state.

How To Run It — Bulk Migration

```
step1_extract_surveys('data/legacy/original_csv/RUSS_24_VALID.CSV')  
step2_upload_surveys('Config', load_config('migration'))  
step3_validate_migration()
```

Steps run one at a time today, so problems can be caught between them.

Historical Migration — Extraction Results

12,578

surveys extracted

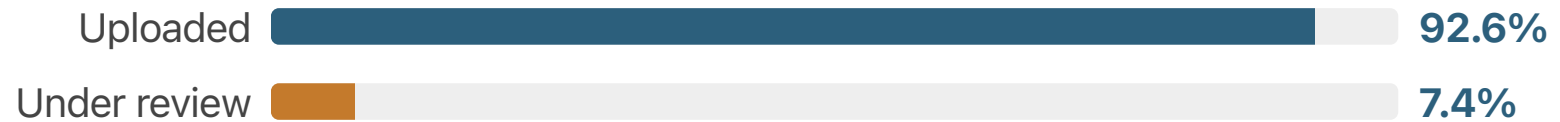
11.65M

records processed

~341 min

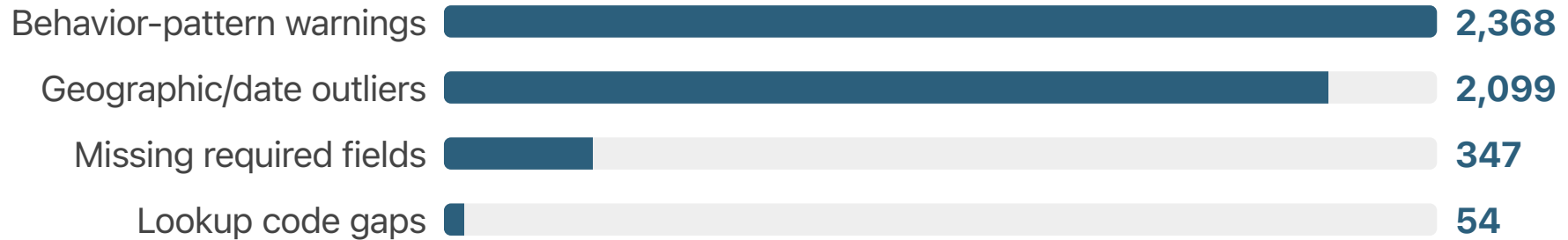
elapsed (~5.7 hrs)

Historical Migration — Validation & Upload Results



11,641 of 12,578 surveys successfully validated and uploaded;
937 still have open issues, pending review.

What's Still Under Review



Counts are validation instances from the review run, not distinct surveys — one survey can trigger more than one.

Validation working as designed — flagging for a human decision, not a tooling problem.

Still To Do

1. **Batch converters** for each contributor's input format
2. **Test the git-based workflow** end to end
3. **Resolve** the remaining 937 surveys
4. **Get an updated data file** from Bob
5. **Transfer the full system to NEAQ** — the big remaining milestone

Timeline

